

API Testing Report

1. Testing Overview

The backend API was tested using **Postman** against the deployed application.

Base URL: `https://nibss-a2rm.vercel.app`

Testing covered authentication, customer access, KYC verification, account management, fund transfers, and transaction retrieval.

2. Authentication Testing

AUTH-001 — Customer Registration

Endpoint: `POST /api/auth/register`

Test Data

```
{
  "name": "Chisom Abraham",
  "email": "customer@example.com",
  "password": "password123"
}
```

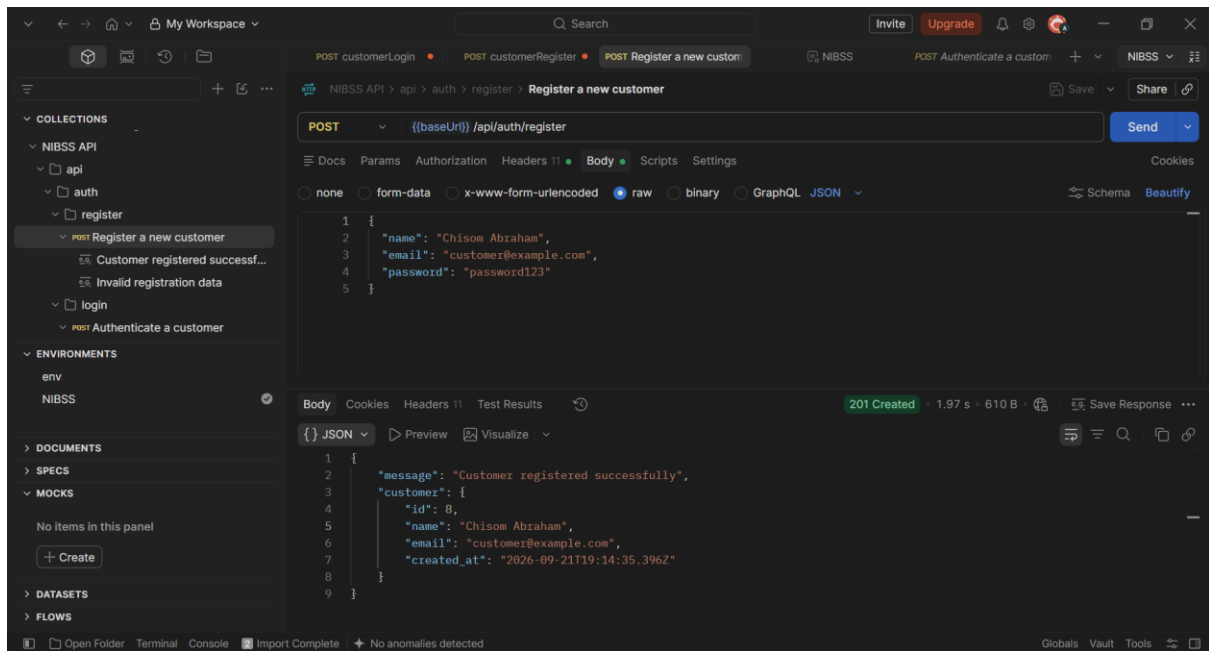
Expected: Customer is registered successfully.

Actual: Customer registered successfully. Customer ID 8 was returned.

Result:  PASS

Screenshot:

[FIGURE 1: AUTH-001 - Customer Registration]



AUTH-002 — Customer Login

Endpoint: POST /api/auth/login

Test Data

```
{
  "email": "customer@example.com",
  "password": "password123"
}
```

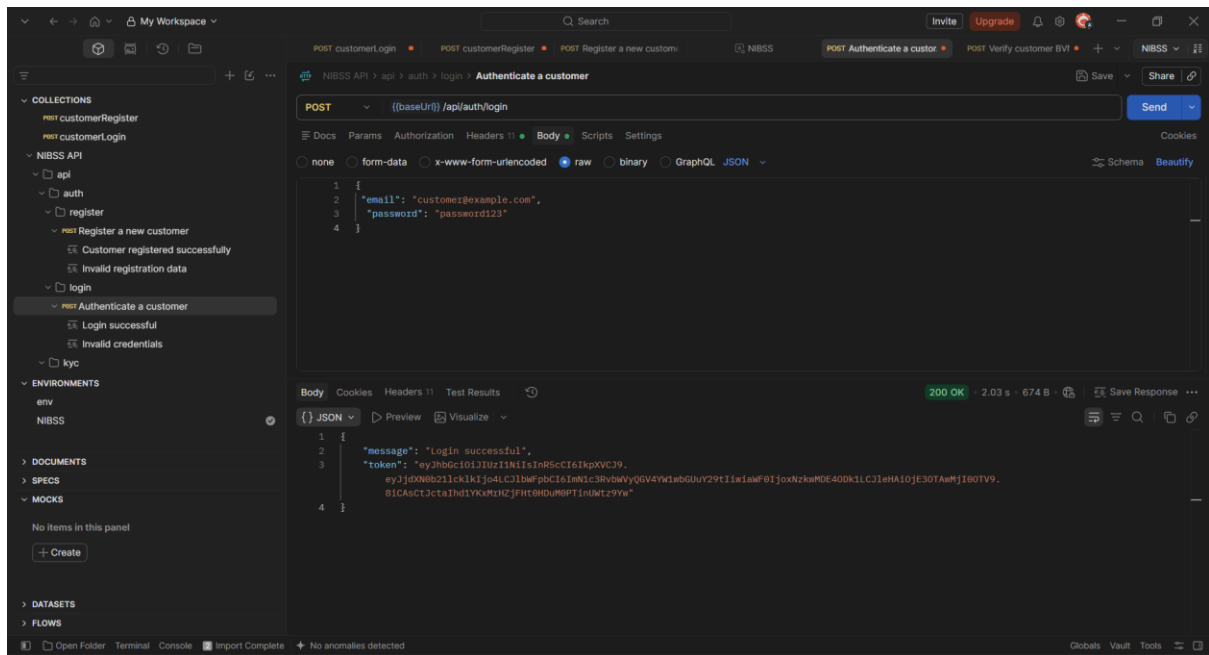
Expected: Login succeeds and an authentication token is returned.

Actual: Login successful and JWT token returned.

Result:  PASS

Screenshot:

[FIGURE 2: AUTH-002 - Customer Login]



3. Customer Testing

CUST-001 — Get Customer Profile

Endpoint: GET /api/customers/me

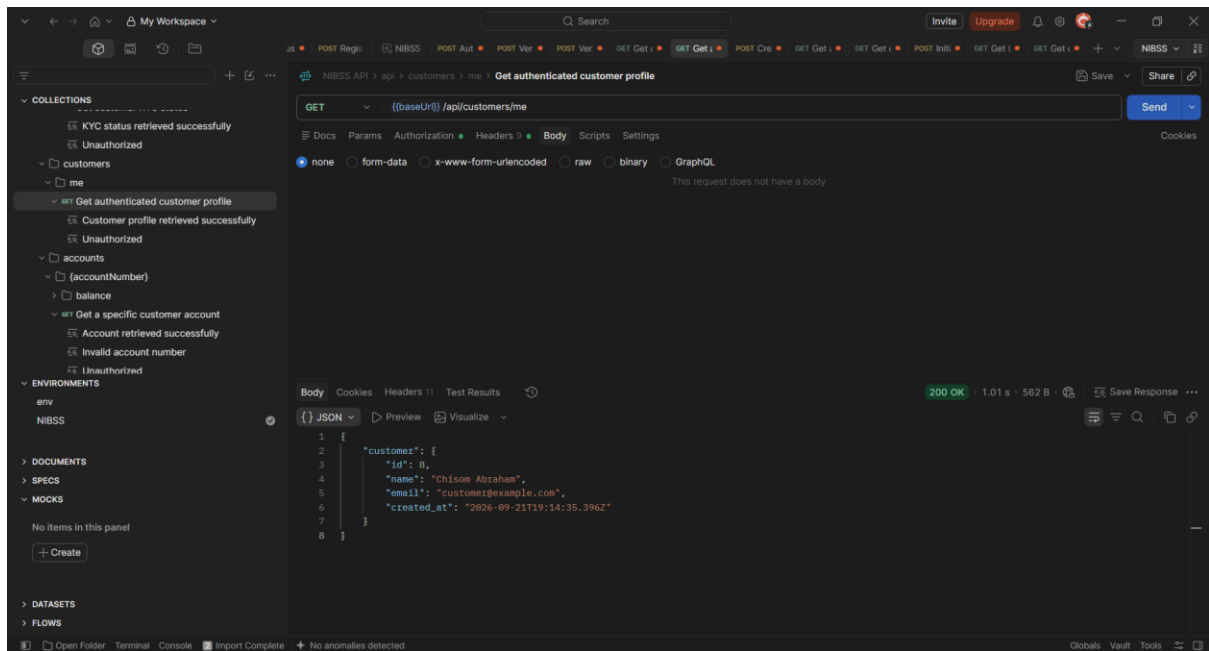
Expected: The authenticated customer's profile is returned.

Actual: Customer ID, name, email, and creation date were returned successfully.

Result: ✔ PASS

Screenshot:

[FIGURE 3: CUST-001 - Customer Profile]



4. KYC Testing

KYC-001 — BVN Verification

Endpoint: POST /api/kyc/bvn

Test Data

```
{
  "bvn": "92647130582",
  "firstName": "Chisom",
  "lastName": "Abraham",
  "dob": "1992-05-15",
  "phone": "08012345618"
}
```

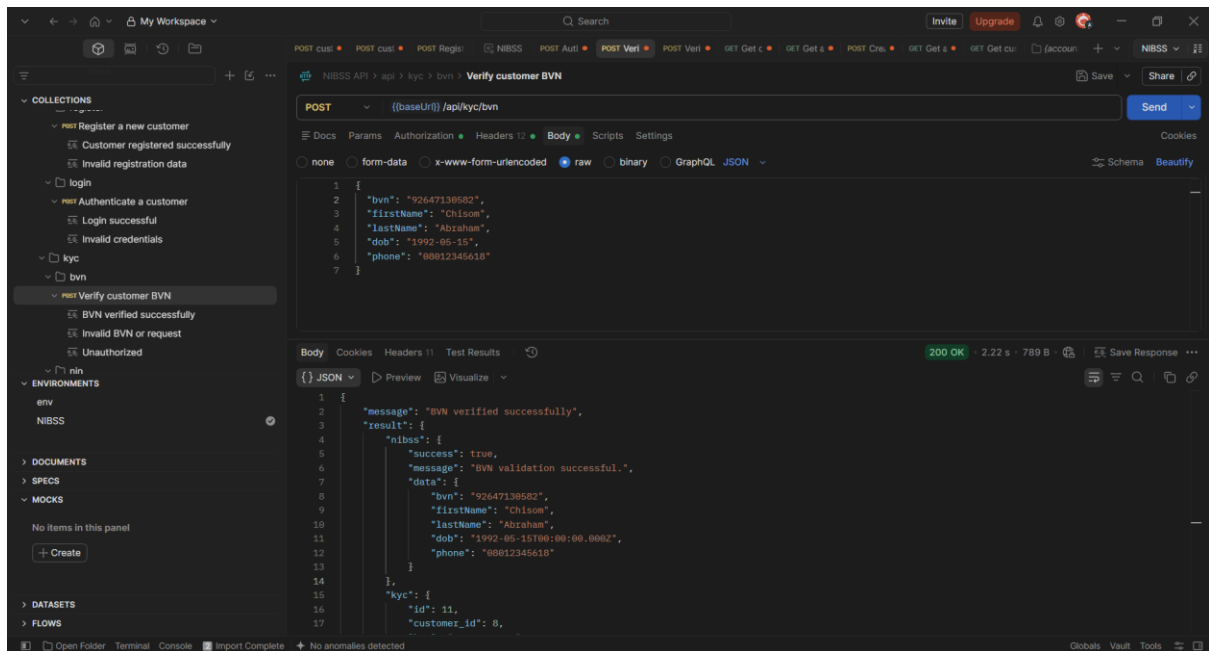
Expected: BVN is verified through the NIBSS integration and the local KYC record is updated.

Actual: BVN verification was successful. The local KYC record shows `bvn_verified: true`.

Result:  PASS

Screenshot:

[FIGURE 4: KYC-001 – BVN Verification]



KYC-002 — NIN Verification

Endpoint: POST /api/kyc/nin

Test Data

```
{
  "nin": "31580947261",
  "firstName": "Chisom",
  "lastName": "Abraham",
  "dob": "1992-05-15"
}
```

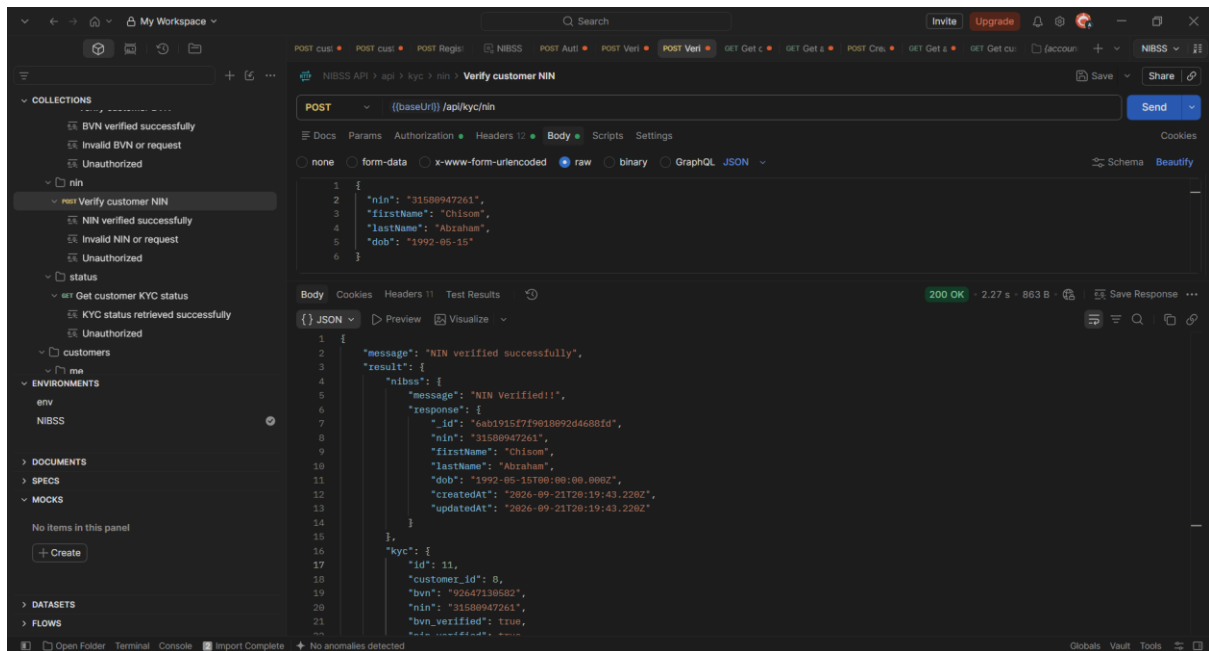
Expected: NIN is verified through the NIBSS integration and the local KYC record is updated.

Actual: NIN verification was successful. The local KYC record shows `nin_verified: true`.

Result: ✔ PASS

Screenshot:

[FIGURE 5: KYC-002 - NIN Verification]



KYC-003 — KYC Status

Endpoint: GET /api/kyc/status

Expected: Current BVN and NIN verification status is returned.

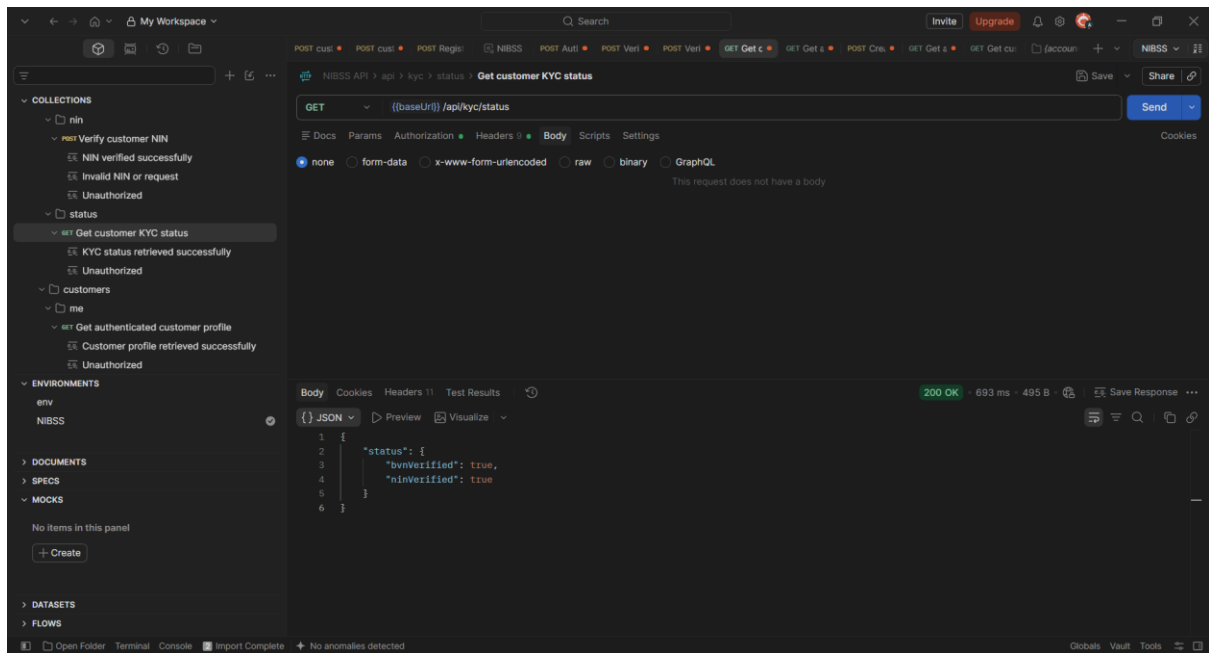
Actual:

```
{
  "bvnVerified": true,
  "ninVerified": true
}
```

Result: ✔ PASS

Screenshot:

[FIGURE 6: KYC-003 - KYC Status]



5. Account Testing

ACCT-001 — Create Account

Endpoint: POST /api/accounts

Test Data

```
{
  "accountType": "savings",
  "kycType": "bvn",
  "kycID": "92647130582",
  "dob": "1992-05-15"
}
```

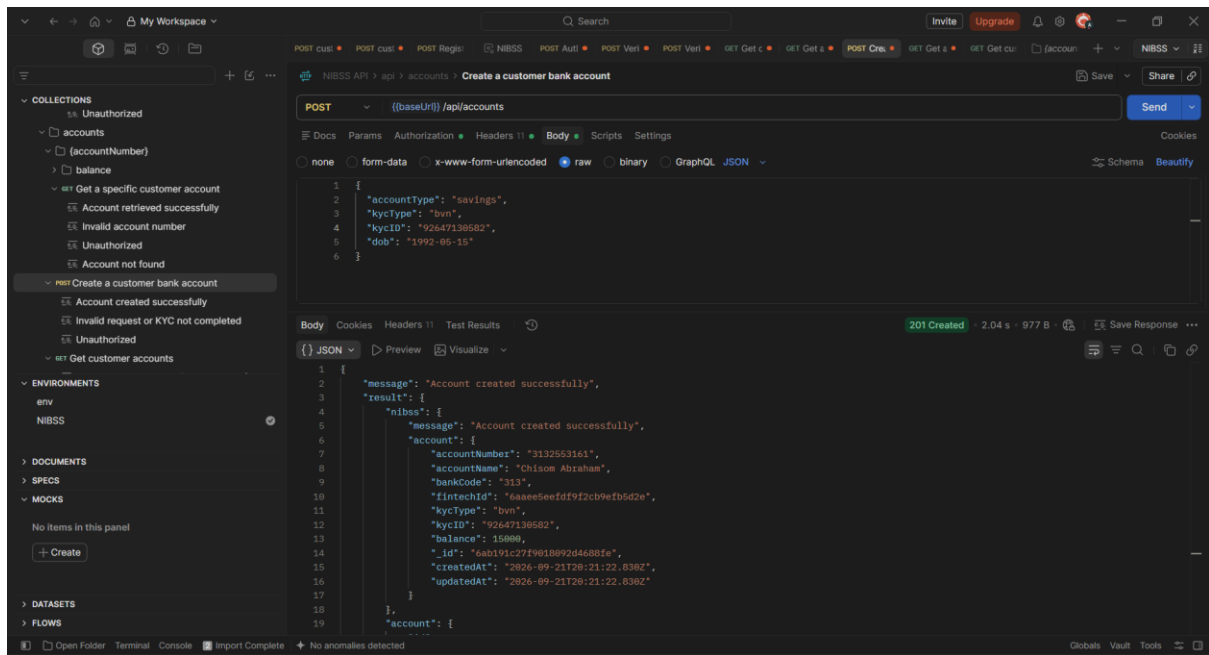
Expected: Account is created through the NIBSS integration and saved locally.

Actual: Account was created successfully. Account number 3132553161 was generated with an initial balance of ₦15,000.

Result:  PASS

Screenshot:

[FIGURE 7: ACCT-001 – Account Creation]



ACCT-002 — Get Customer Accounts

Endpoint: GET `/api/accounts`

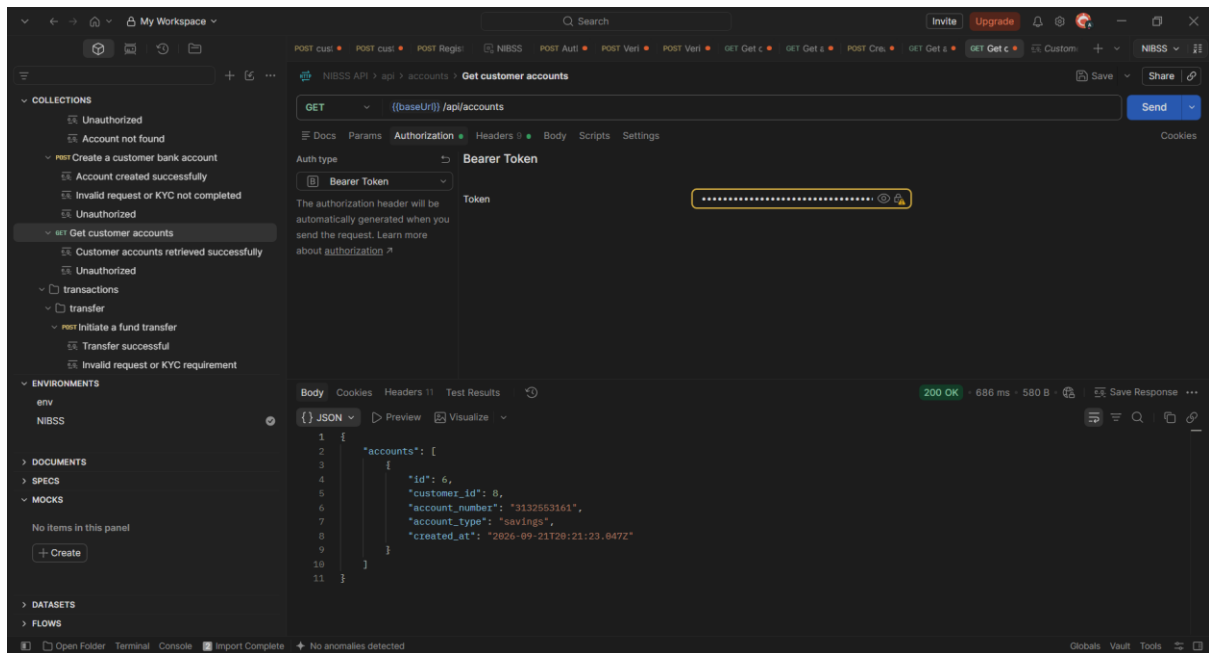
Expected: Accounts belonging to the authenticated customer are returned.

Actual: The newly created savings account was returned successfully.

Result: ✔ PASS

Screenshot:

[FIGURE 8: ACCT-002 - Customer Accounts]



ACCT-003 — Get Account by Account Number

Endpoint: GET `/api/accounts/:accountNumber`

Test: Account number 3132553161

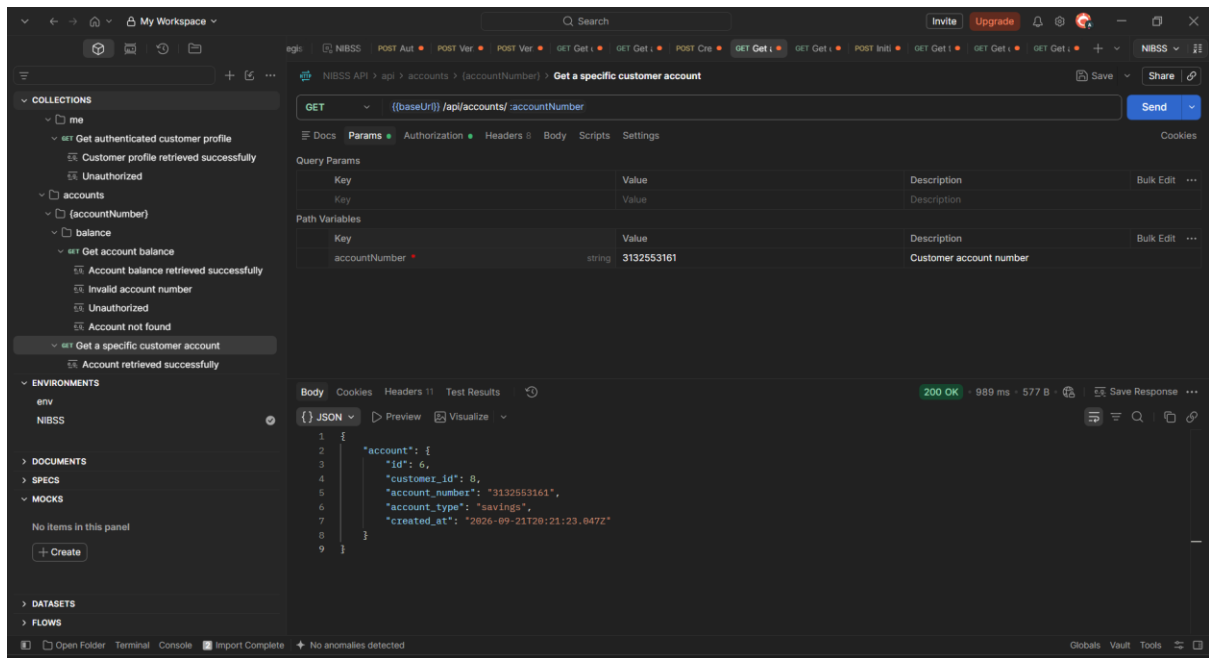
Expected: The requested account belonging to the authenticated customer is returned.

Actual: The account details were returned successfully, including the account number and account type.

Result: ✔ PASS

Screenshot:

[FIGURE 9: ACCT-003 - Get Account by Account Number]



ACCT-004 — Get Account Balance

Endpoint: GET `/api/accounts/:accountNumber/balance`

Test: Account number 3132553161

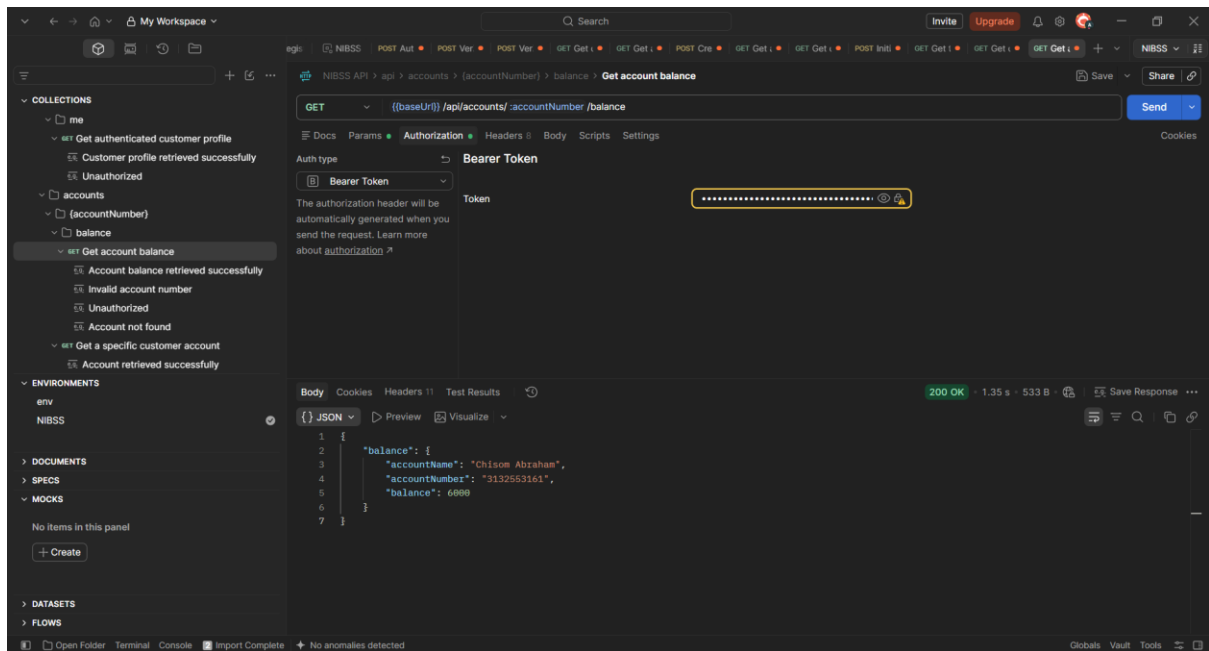
Expected: The current account balance is returned.

Actual: The account balance was returned successfully. The balance was ₦6,000.

Result:  PASS

Screenshot:

[FIGURE 10: ACCT-004 - Account Balance]



6. Transfer Testing

TRANS-001 — Initiate Transfer

Endpoint: POST /api/transactions/transfer

Test Data

```
{
  "fromAccount": "3132553161",
  "toAccount": "3135379018",
  "amount": 9000
}
```

Expected: Transfer is processed through NIBSS and a transaction record is created.

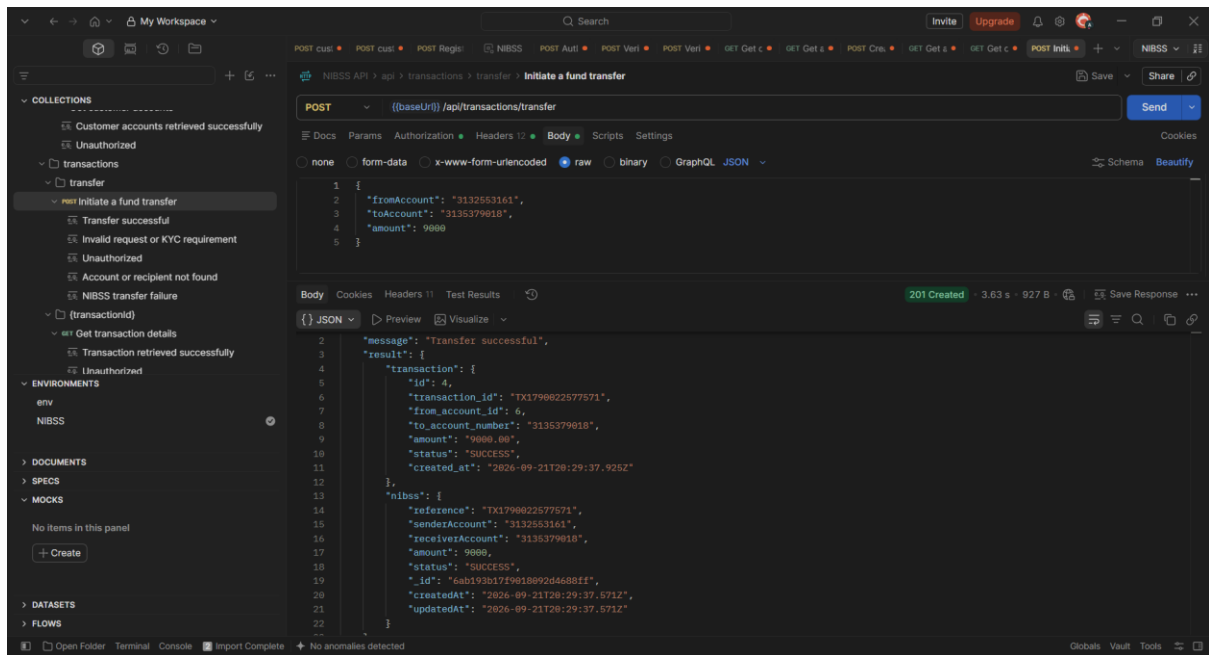
Actual: The ₦9,000 transfer was completed successfully. Both the NIBSS response and local transaction record returned status `SUCCESS`.

Transaction ID: TX1790022577571

Result: ✔ PASS

Screenshot:

[FIGURE 11: TRANS-001 - Successful Transfer]



7. Transaction Testing

TXN-001 — Get Transaction Details

Endpoint: GET `/api/transactions/:transactionId`

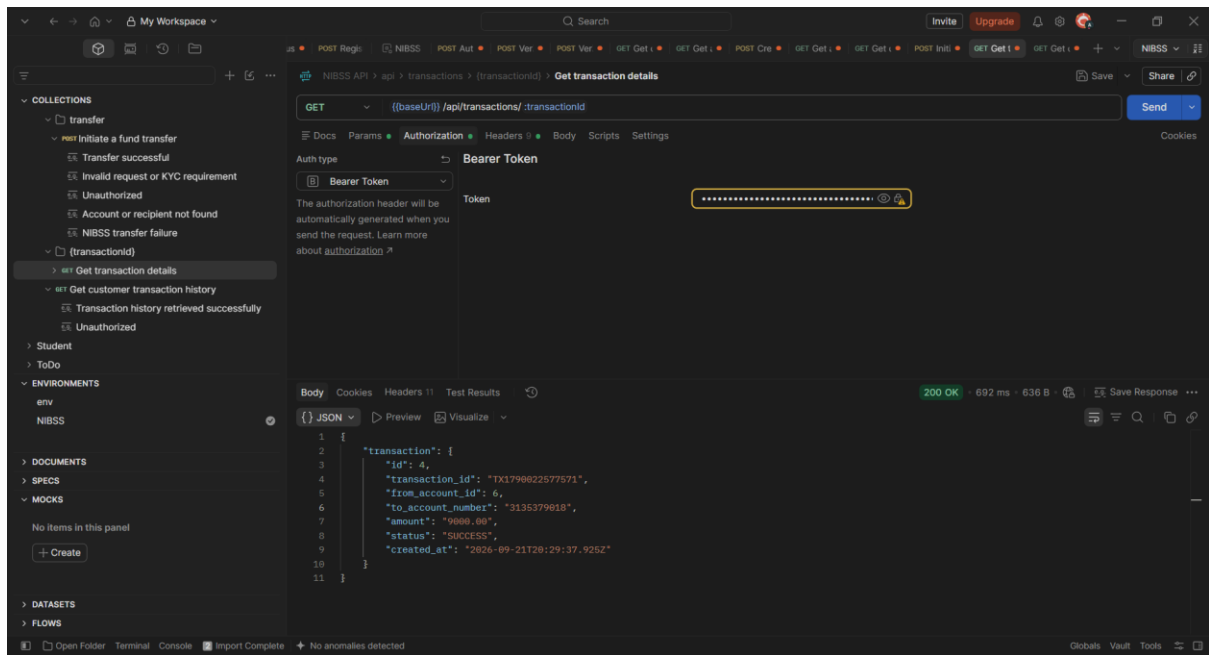
Expected: The requested transaction is returned.

Actual: Transaction details were returned successfully, including the transaction ID, amount, recipient account, status, and creation time.

Result: ✔ PASS

Screenshot:

[FIGURE 12: TXN-001 - Transaction Details]



TXN-002 — Get Transaction History

Endpoint: GET /api/transactions

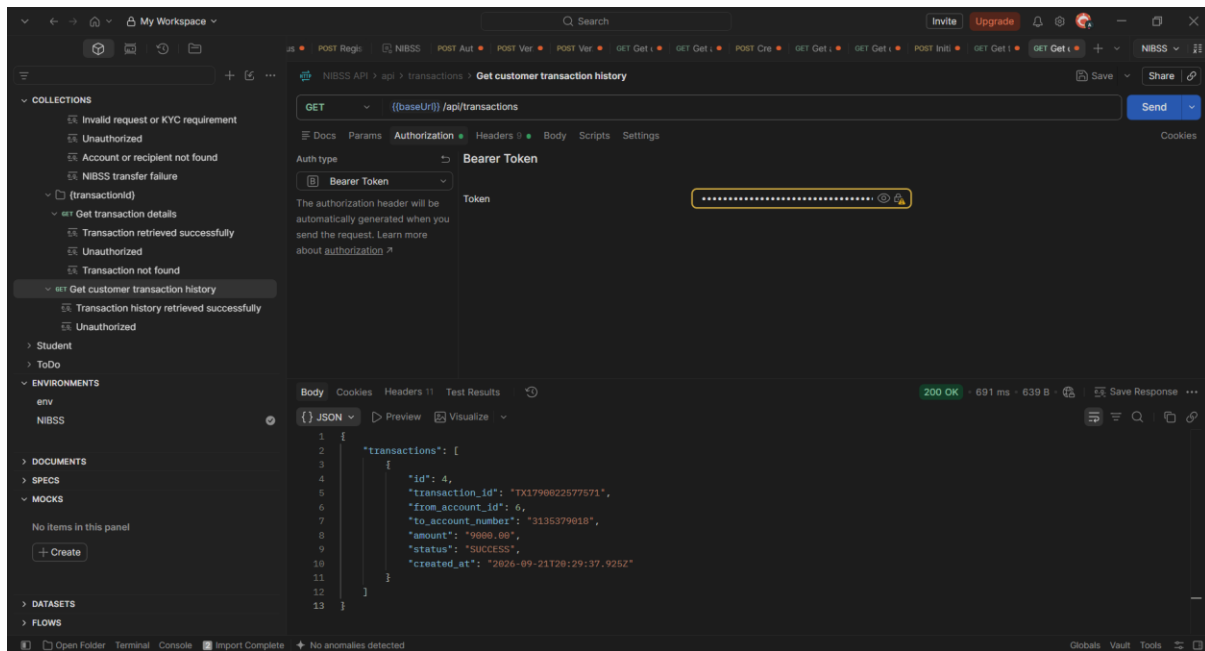
Expected: Transactions belonging to the authenticated customer are returned.

Actual: The completed ₦9,000 transaction was returned successfully.

Result: ✔ PASS

Screenshot:

[FIGURE 13: TXN-002 - Transaction History]



8. Test Summary

Area	Tests	Passed
Authentication	2	2
Customer	1	1
KYC	3	3
Accounts	4	4
Transfers	1	1
Transactions	2	2
Total	13	13

Overall result: 13/13 documented functional tests passed.

The testing confirmed that the backend successfully handles customer authentication, customer profile access, KYC verification, account creation and retrieval, balance retrieval, fund transfers, and transaction retrieval.

Testing evidence: A Postman screenshot is provided for each test case above.

9. Final Testing Checklist

- Customer registration tested
- Customer login tested
- Customer profile tested
- BVN verification tested

- NIN verification tested
- KYC status tested
- Account creation tested
- Customer accounts retrieval tested
- Individual account retrieval tested
- Account balance tested
- Successful transfer tested
- Transaction details tested
- Transaction history tested
- Postman screenshots captured